

Task 2: Exploratory Data Analysis & Business Intelligence

BigQuery SQL Analysis — ApexPlanet Software Pvt. Ltd.

Data Analytics Internship Deliverable

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Executive Summary

This report defines the core business questions for exploratory analysis of the ApexPlanet Sales_Dataset, now hosted in BigQuery (aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset), and provides the production-ready SQL to answer each one. Seven questions were selected to cover revenue trends, product and category performance, regional performance, customer demographics, customer value/retention, and category growth momentum. Each query uses CTEs, window functions, and BigQuery formatting functions to produce stakeholder-ready output directly from the warehouse. A final section specifies exactly which columns to export for the Python EDA, statistical analysis, and visualization phase.

Step 1 — Business Questions

Before writing any SQL, the following questions were defined to anchor the analysis in decisions the business actually needs to make — not just numbers that are easy to compute.

#	Business Question	Why It Matters
1	How does revenue trend month over month and year over year?	Reveals growth trajectory and seasonality, and is the baseline metric every stakeholder review starts with.
2	Which products generate the most revenue?	Identifies which SKUs to prioritize for stock, promotion, and supplier negotiation.
3	Which product categories perform best, and what share of revenue does each hold?	Guides category-level budget allocation and cross-sell/bundling strategy.
4	Which cities/regions drive the most revenue and orders?	Informs regional marketing spend, warehouse placement, and expansion priorities.
5	How does purchasing behavior differ by age group and gender?	Enables targeted marketing creative and product recommendations by demographic segment.
6	Who are the highest-value customers, and how many are repeat buyers?	Surfaces candidates for a loyalty program and quantifies retention vs. one-time acquisition.
7	Which categories are growing or declining month over month?	Flags emerging or fading demand early enough to act on it, rather than reading it after a quarter closes.

Step 2 — SQL Query Documentation

Each query below is written in BigQuery Standard SQL, targeting the `cleaned_sales_dataset` table directly. All queries are read-only (SELECT), safe to run repeatedly, and use `ROUND()` / `FORMAT_DATE()` for stakeholder-readable output.

Query 1: Monthly Revenue Trend

Business Question: How does revenue trend month over month and year over year?

```
WITH monthly_sales AS (
  SELECT
    EXTRACT(YEAR FROM Order_Date) AS Order_Year,
    EXTRACT(MONTH FROM Order_Date) AS Order_Month,
    DATE_TRUNC(Order_Date, MONTH) AS Sales_Month,
    SUM(Total_Sales) AS Total_Revenue,
    COUNT(DISTINCT Order_ID) AS Total_Orders
  FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
  GROUP BY Order_Year, Order_Month, Sales_Month
)
SELECT
  Order_Year,
  Order_Month,
  FORMAT_DATE('%B %Y', Sales_Month) AS Month_Label,
  ROUND(Total_Revenue, 2) AS Total_Revenue,
  Total_Orders,
  ROUND(Total_Revenue / Total_Orders, 2) AS Avg_Order_Value
FROM monthly_sales
ORDER BY Order_Year, Order_Month;
```

Row	Order_Year	Order_Month	Month_Label	Total_Revenue	Total_Orders	Avg_Order_Value
1	2025	1	January 2025	10096190.73	76	132844.61
2	2025	2	February 2025	11511181.46	86	133850.95
3	2025	3	March 2025	13059899.94	89	146740.45
4	2025	4	April 2025	12222700.17	80	152783.75
5	2025	5	May 2025	10984689.25	85	129231.64
6	2025	6	June 2025	12912332.64	83	155570.27
7	2025	7	July 2025	11746226.8	83	141520.8
8	2025	8	August 2025	9448471.22	75	125979.62
9	2025	9	September 2025	9179896.29	79	116201.22

10	2025	10	October 2025	12500936.5	84	148820.67
11	2025	11	November 2025	12627620.22	77	163995.07
12	2025	12	December 2025	12299904.44	98	125509.23
13	2026	1	January 2026	809389.99	5	161878.0

Insight: Revenue remained stable throughout 2025, with **March recording the highest sales** and **September the lowest among full months**, indicating moderate seasonal variation. The low revenue in **January 2026** is likely due to incomplete monthly data rather than a genuine decline in business performance.

Query 2: Top 10 Best-Selling Products

Business Question: Which products generate the most revenue? `LIMIT 10;`

```
SELECT
  Product,
  Category,
  ROUND(SUM(Total_Sales), 2) AS Total_Revenue,
  SUM(Quantity) AS Units_Sold,
  COUNT(DISTINCT Order_ID) AS Order_Count,
  ROUND(SUM(Total_Sales) / COUNT(DISTINCT Order_ID), 2) AS Avg_Order_Value
FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
GROUP BY Product, Category
ORDER BY Total_Revenue DESC
LIMIT 10;
```

Row	Product	Category	Total_Revenue	Units_Sold	Order_Count	Avg_Order_Value
1	Laptop	Electronics	25443008.51	970	170	149664.76
2	Mobile	Electronics	25335573.19	1008	184	137693.33
3	Book	Education	25031689.4	977	178	140627.47
4	Rice	Grocery	22231711.28	826	153	145305.3
5	Chair	Furniture	21521561.48	855	159	135355.73
6	Shoes	Fashion	19835895.79	799	156	127153.18

Insight: **Laptop and Mobile** were the highest revenue-generating products, with Electronics dominating the top-selling list. This indicates strong customer demand for electronic products and highlights them as the most valuable category for inventory planning and future promotional campaigns.

Query 3: Category Performance Overview

Business Question: Which product categories perform best, and what share of revenue does each hold?

```
WITH category_summary AS (
  SELECT
    Category,
    SUM(Total_Sales) AS Total_Revenue,
    SUM(Quantity) AS Units_Sold,
    COUNT(DISTINCT Order_ID) AS Order_Count,
    ROUND(AVG(Unit_Price), 2) AS Avg_Unit_Price
  FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
  GROUP BY Category
)
SELECT
  Category,
  ROUND(Total_Revenue,2) AS Total_Revenue,
  Units_Sold,
  Order_Count,
  Avg_Unit_Price,
  ROUND(Total_Revenue / SUM(Total_Revenue) OVER (), 4) AS Pct_Of_Total_Revenue
FROM category_summary
ORDER BY Total_Revenue DESC;
```

Row	Category	Total_Revenue	Units_Sold	Order_Count	Avg_Unit_Price	Pct_Of_Total_Revenue
1	Electronics	50778581.7	1978	354	25165.39	0.3643
2	Education	25031689.4	977	178	24746.71	0.1796
3	Grocery	22231711.28	826	153	26360.24	0.1595
4	Furniture	21521561.48	855	159	25788.9	0.1544
5	Fashion	19835895.79	799	156	25895.97	0.1423

Insight: Electronics contributed over one-third of total revenue, making it the strongest-performing category in terms of sales, units sold, and order volume. The remaining categories showed relatively balanced contributions, indicating a diversified product portfolio with Electronics as the primary revenue

Query 4: City-Wise Sales Performance

Business Question: Which cities/regions drive the most revenue and orders?

```
SELECT
  City,
  ROUND(SUM(Total_Sales), 2) AS Total_Revenue,
  COUNT(DISTINCT Order_ID) AS Total_Orders,
  COUNT(DISTINCT Customer_ID) AS Unique_Customers,
  ROUND(SUM(Total_Sales) / COUNT(DISTINCT Order_ID), 2) AS Avg_Order_Value
FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
GROUP BY City
ORDER BY Total_Revenue DESC;
```

Row	City	Total_Revenue	Total_Orders	Unique_Customers	Avg_Order_Value
1	Patna	19285966.89	135	134	142859.01
2	Kolkata	18884349.57	133	132	141987.59
3	Bengaluru	18773574.32	122	120	153881.76
4	Mumbai	18757050.17	131	130	143183.59
5	Hyderabad	17166766.87	125	125	137334.13
6	Delhi	16097079.0	125	124	128776.63
7	Pune	14513175.9	99	99	146597.74
8	Gaya	14380859.39	117	117	122913.33
9	Unknown	1540617.54	13	13	118509.04

Insight: Patna generated the highest revenue, followed closely by Kolkata, Bengaluru, and Mumbai, showing that sales are well distributed across major cities. This geographical diversity reduces dependence on a single market while identifying key regions for future business expansion.

Query 5: Customer Demographics — Age Group & Gender

Business Question: How does purchasing behavior differ by age group and gender?

```

WITH demographics AS (
  SELECT
    CASE
      WHEN Age BETWEEN 18 AND 25 THEN '18-25'
      WHEN Age BETWEEN 26 AND 35 THEN '26-35'
      WHEN Age BETWEEN 36 AND 45 THEN '36-45'
      WHEN Age BETWEEN 46 AND 55 THEN '46-55'
      WHEN Age BETWEEN 56 AND 65 THEN '56-65'
      ELSE 'Other'
    END AS Age_Group,
    Gender,
    Order_ID,
    Total_Sales
  FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
)
SELECT
  Age_Group,
  Gender,
  ROUND(SUM(Total_Sales), 2) AS Total_Revenue,
  COUNT(DISTINCT Order_ID) AS Order_Count,
  ROUND(SUM(Total_Sales) / COUNT(DISTINCT Order_ID), 2) AS Avg_Order_Value
FROM demographics
GROUP BY Age_Group, Gender
ORDER BY Age_Group, Gender;

```

Row	Age_Group	Gender	Total_Revenue	Order_Count	Avg_Order_Value
1	18-25	Female	11248328.82	71	158427.17
2	18-25	Male	11035768.12	87	126847.91
3	26-35	Female	10927235.18	98	111502.4
4	26-35	Male	18705101.78	121	154587.62
5	36-45	Female	17088042.98	124	137806.8
6	36-45	Male	14445202.73	99	145911.14
7	46-55	Female	13009983.23	91	142966.85
8	46-55	Male	15206525.4	111	136995.72
9	56-65	Female	14662298.81	105	139640.94
10	56-65	Male	13070952.6	93	140547.88

Insight: Customers aged **26–35 years**, particularly males, generated the highest revenue, making them the most valuable customer segment. Strong contributions from the 36–45 age group also suggest that middle-aged customers represent an important target audience for marketing strategies.

Query 6: High-Value & Repeat Customers

Business Question: Who are the highest-value customers, and how many are repeat buyers?

```

WITH customer_summary AS (
  SELECT
    Customer_ID,
    Customer_Name,
    COUNT(DISTINCT Order_ID) AS Order_Count,
    SUM(Total_Sales) AS Total_Spend,
    ROUND(SUM(Total_Sales) / COUNT(DISTINCT Order_ID), 2) AS Avg_Order_Value
  FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
  GROUP BY Customer_ID, Customer_Name
)
SELECT
  Customer_ID,
  Customer_Name,
  Order_Count,
  Total_Spend,
  Avg_Order_Value,
  CASE WHEN Order_Count > 1 THEN 'Repeat Customer' ELSE 'One-Time Customer' END AS Customer_Type
FROM customer_summary
ORDER BY Total_Spend DESC
LIMIT 10;

```

Row	Customer_ID	Customer_Name	Order_Count	Total_Spend	Avg_Order_Value	Customer_Type
1	CUST2062	Customer_254	1	493677.5	493677.5	One-Time Customer
2	CUST4706	Customer_200	1	492174.1	492174.1	One-Time Customer
3	CUST1711	Customer_36	1	490866.4	490866.4	One-Time Customer
4	CUST4869	Customer_392	1	485667.8	485667.8	One-Time Customer
5	CUST7416	Customer_191	1	482551.9	482551.9	One-Time Customer
6	CUST9551	Customer_230	1	474935.9	474935.9	One-Time Customer
7	CUST8381	Customer_112	1	473144.2	473144.2	One-Time Customer
8	CUST4037	Customer_205	1	470798.6	470798.6	One-Time Customer
9	CUST4003	Customer_305	1	468522.0	468522.0	One-Time Customer
10	CUST4584	Customer_138	1	462701.3	462701.3	One-Time Customer

Insight: The highest-spending customers each placed only **one order**, indicating a lack of repeat purchases in the dataset. This highlights an opportunity to improve customer retention through loyalty programs and personalized engagement strategies to increase long-term customer value.

Query 7: Month-over-Month Category Growth Rate

Business Question: Which categories are growing or declining month over month?

```

WITH monthly_category_sales AS (
  SELECT
    Category,
    DATE_TRUNC(Order_Date, MONTH) AS Sales_Month,
    SUM(Total_Sales) AS Monthly_Revenue
  FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
  GROUP BY Category, Sales_Month
),
growth AS (
  SELECT
    Category,
    Sales_Month,
    Monthly_Revenue,
    LAG(Monthly_Revenue) OVER (PARTITION BY Category ORDER BY Sales_Month) AS Prev_Month_Revenue
  FROM monthly_category_sales
)
SELECT
  Category,
  FORMAT_DATE('%B %Y', Sales_Month) AS Month_Label,

```

```

ROUND(Monthly_Revenue, 2) AS Monthly_Revenue,
ROUND(Prev_Month_Revenue, 2) AS Prev_Month_Revenue,
ROUND(SAFE_DIVIDE(Monthly_Revenue - Prev_Month_Revenue, Prev_Month_Revenue) * 100, 2) AS
MoM_Growth_Pct
FROM growth
ORDER BY Category, Sales_Month;

```

Row	Category	Month_Label	Monthly_Revenue	Perv_Month_Revenue	MoM_Growth_Pct
1	Education	January 2025	2191833.62	null	null
2	Education	February 2025	1721264.51	2191833.62	-21.47
3	Education	March 2025	1497961.62	1721264.51	-12.97
4	Education	April 2025	2450371.44	1497961.62	63.58
5	Education	May 2025	1578839.04	2450371.44	-35.57
6	Education	June 2025	2851586.64	1578839.04	80.61
7	Education	July 2025	1970820.43	2851586.64	-30.89
8	Education	August 2025	1275407.38	1970820.43	-35.29
9	Education	September 2025	2800843.84	1275407.38	119.6
10	Education	October 2025	2252314.83	2800843.84	-19.58
11	Education	November 2025	1883866.76	2252314.83	-16.36
12	Education	December 2025	2556579.29	1883866.76	35.71
13	Electronics	January 2025	3990715.6	null	null
14	Electronics	February 2025	4723355.96	3990715.6	18.36
15	Electronics	March 2025	4802723.0	4723355.96	1.68
16	Electronics	April 2025	4809595.36	4802723.0	0.14
17	Electronics	May 2025	4658213.31	4809595.36	-3.15
18	Electronics	June 2025	4847273.99	4658213.31	4.06

19	Electronics	July 2025	3754882.46	4847273.99	-22.54
20	Electronics	August 2025	2927508.76	3754882.46	-22.03
21	Electronics	September 2025	2224504.0	2927508.76	-24.01
22	Electronics	October 2025	3406542.61	2224504.0	53.14
23	Electronics	November 2025	5735227.11	3406542.61	68.36
24	Electronics	December 2025	4799958.62	5735227.11	-16.31
25	Electronics	January 2026	98080.92	4799958.62	-97.96
26	Fashion	January 2025	997172.84	null	null
27	Fashion	February 2025	1384993.02	997172.84	38.89
28	Fashion	March 2025	1629558.18	1384993.02	17.66
29	Fashion	April 2025	2267590.93	1629558.18	39.15
30	Fashion	May 2025	1471420.14	2267590.93	-35.11
31	Fashion	June 2025	935768.65	1471420.14	-36.4
32	Fashion	July 2025	2145178.66	935768.65	129.24
33	Fashion	August 2025	2407650.93	2145178.66	12.24
34	Fashion	September 2025	1622218.31	2407650.93	-32.62
35	Fashion	October 2025	1639391.61	1622218.31	1.06
36	Fashion	November 2025	1521347.47	1639391.61	-7.2
37	Fashion	December 2025	1466446.52	1521347.47	-3.61
38	Fashion	January 2026	347158.53	1466446.52	-76.33
39	Furniture	January 2025	1692072.45	null	null
40	Furniture	February 2025	1734334.31	1692072.45	2.5

41	Furniture	March 2025	2824297.69	1734334.31	62.85
42	Furniture	April 2025	887339.59	2824297.69	-68.58
43	Furniture	May 2025	1960602.17	887339.59	120.95
44	Furniture	June 2025	861962.82	1960602.17	-56.04
45	Furniture	July 2025	2817541.9	861962.82	226.88
46	Furniture	August 2025	1547712.91	2817541.9	-45.07
47	Furniture	September 2025	1835434.18	1547712.91	18.59
48	Furniture	October 2025	3104130.73	1835434.18	69.12
49	Furniture	November 2025	791070.79	3104130.73	-74.52
50	Furniture	December 2025	1465061.94	791070.79	85.2

Insight: Category performance fluctuated across months, with **Electronics showing strong growth during October and November** before declining in January 2026 due to partial data. Monitoring monthly category trends can support better inventory planning and seasonal marketing decisions.

Data for Python EDA, Statistical Analysis & Visualization

Rather than exporting the raw table as-is, run the query below in BigQuery and export the result (Save Results → CSV / Google Sheets, or bq extract) — it produces one flat, analysis-ready table with Age_Group and date parts already derived, so no re-cleaning is needed in Python.

Export Query

```
SELECT
  Order_ID,
  Order_Date,
  Customer_ID,
  Customer_Name,
  Age,
  CASE
    WHEN Age BETWEEN 18 AND 25 THEN '18-25'
    WHEN Age BETWEEN 26 AND 35 THEN '26-35'
    WHEN Age BETWEEN 36 AND 45 THEN '36-45'
    WHEN Age BETWEEN 46 AND 55 THEN '46-55'
    WHEN Age BETWEEN 56 AND 65 THEN '56-65'
    ELSE 'Other'
  END AS Age_Group,
  Gender,
  City,
  Product,
  Category,
  Quantity,
  Unit_Price,
  Total_Sales,
  EXTRACT(YEAR FROM Order_Date) AS Order_Year,
  EXTRACT(MONTH FROM Order_Date) AS Order_Month,
  FORMAT_DATE('%B', Order_Date) AS Order_Month_Name
FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
ORDER BY Order_Date;
```

What Each Analysis Phase Needs

Analysis Type	Columns Needed	Purpose
Statistical / descriptive analysis	Age, Quantity, Unit_Price, Total_Sales	Mean, median, std dev, correlation, distribution checks
Segmentation analysis	Gender, City, Category, Age_Group	Group comparisons, ANOVA/chi-square, cross-tab visuals
Time-series / seasonality	Order_Date, Order_Year, Order_Month, Order_Month_Name	Trend lines, month-over-month and year-over-year comparisons
Customer-level analysis	Customer_ID, Order_ID, Total_Sales	Repeat-purchase rate, customer lifetime value, RFM scoring

In Python, load this export directly with `pandas.read_csv()` (or `pandas-gbq` to query BigQuery live) and it will already contain every field needed for descriptive statistics, correlation analysis, segmentation, and time-series visualization without additional joins.

Conclusion

These seven queries turn the cleaned Task 1 dataset into stakeholder-ready answers on growth, product mix, geography, demographics, and customer value — and the export query hands off a single clean table for the Python EDA phase. Together they form the SQL foundation for the BI dashboard and statistical analysis to follow in later tasks.